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COSC320-001

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Project 1 Read Me

General:

This project took awhile. I am pretty sure the 2D regression line I computed is correct, however I am unsure about the 3D regression.

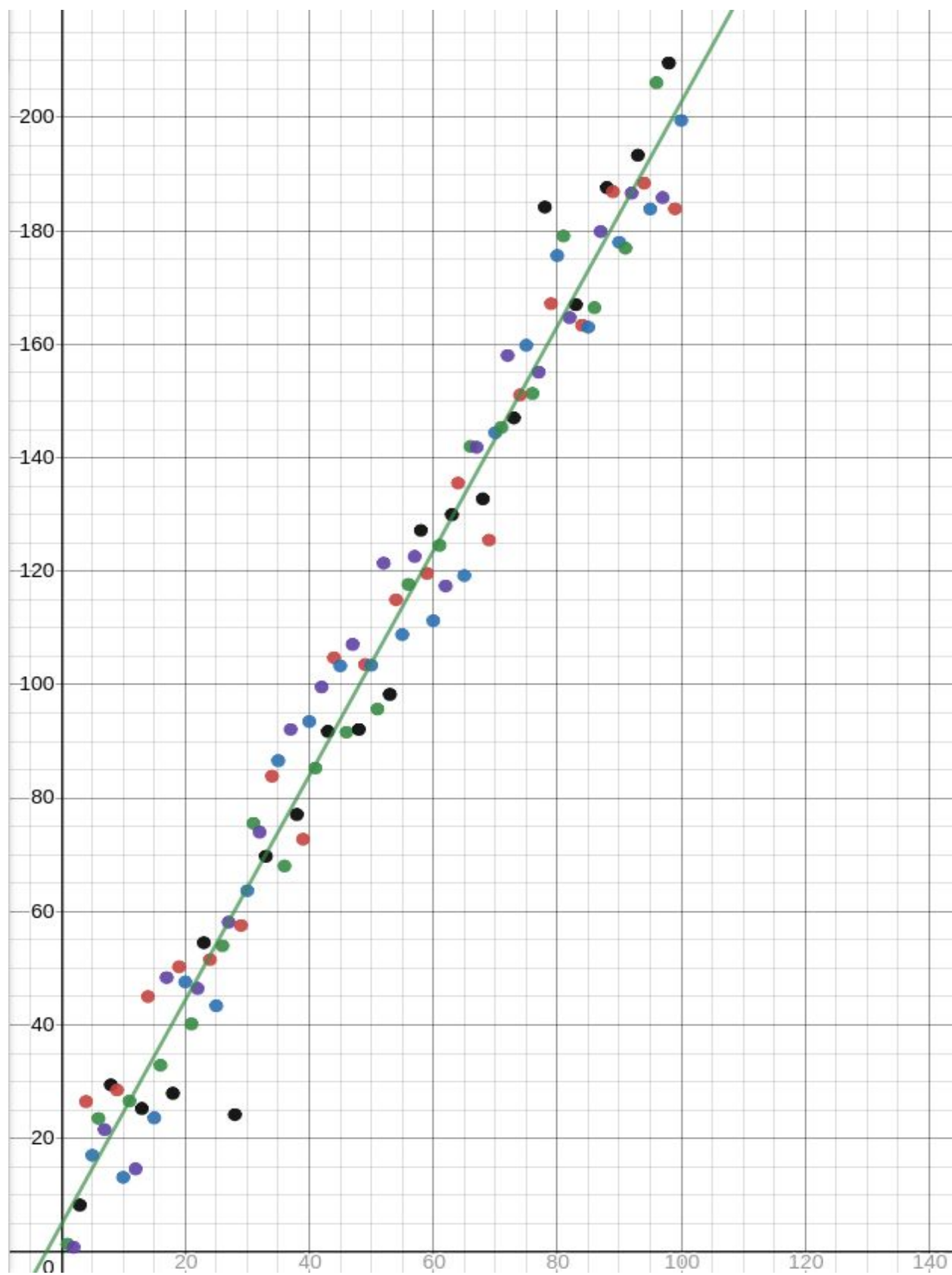
There is a function called Determinant which is for the **Non-singular Extra Credit**.

In the regression exec file there is a method that allows you to do the **2D** regression for a select number of data values given a file name.

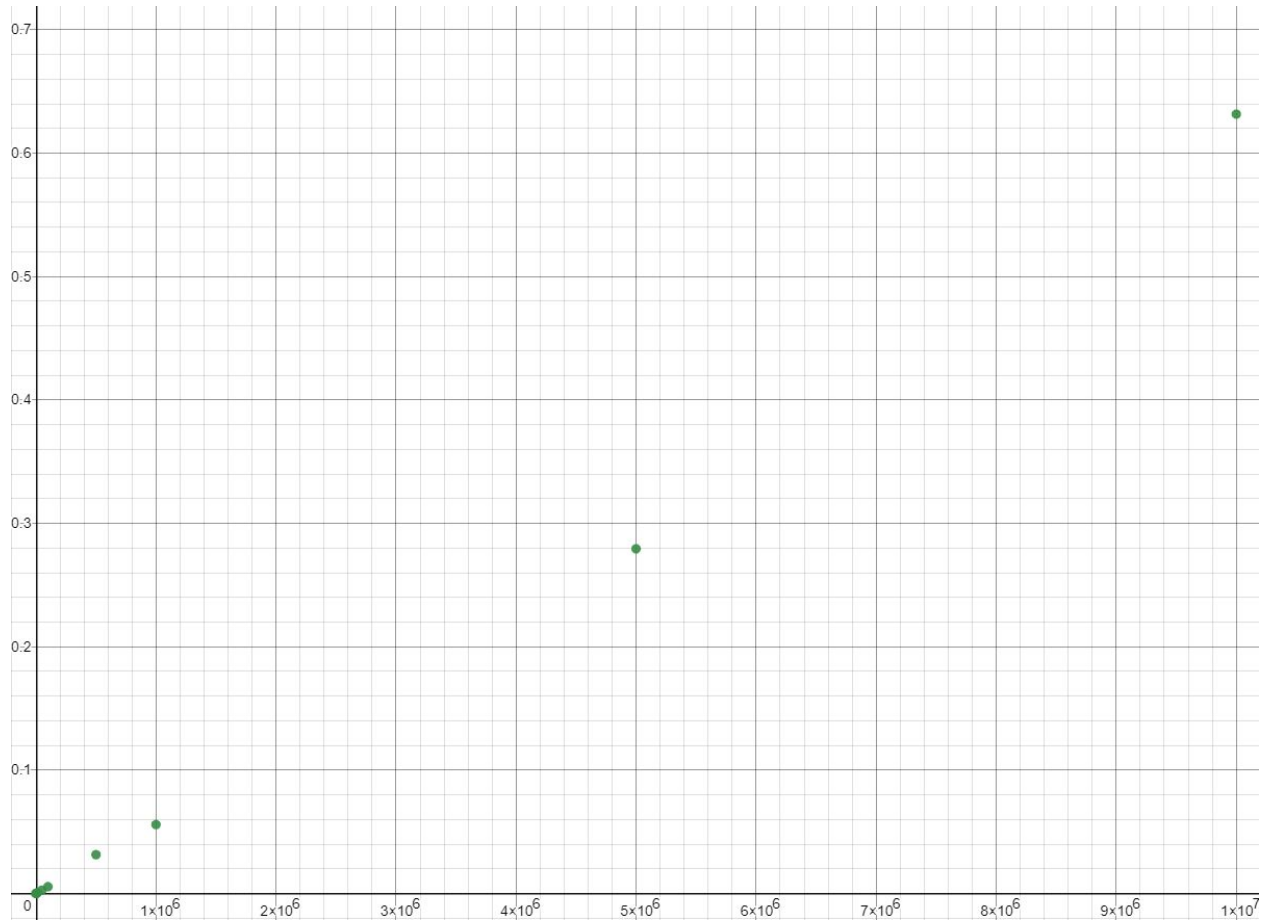
Time Complexity:

The time gathered when computing $\hat{\beta} = (A^T * A)^{-1} * A^T * b$ seems to be extremely low, as there are not many operations needed. If I had to guess the big O, it would probably be $\sim O(\log(n))$ or less.

SEE REGRESSION LINE COMPUTED ON NEXT PAGE



*2D Regression Line



*Time Complexity for different data sizes.

Data Value Size	A * Beta Time
100	0.0000108
1000	0.0000604
5000	0.0002765
10000	0.0005482
50000	0.0028164
100000	0.0055774
500000	0.0315412
1000000	0.055937
5000000	0.279369
10000000	0.631597